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Step 7 - Physical Timer

Reference Note

This chapter is based on the ARM Generic Timer Architecture.

The Physical Timer generates events using the Physical Counter.

Learning Objectives

After this chapter you should understand:

- Purpose of the Physical Timer
- Key Physical Timer registers
- Compare-based timer operation
- Timer expiry concept

1. Physical Timer Architecture

Component	Purpose
CNTPCT_EL0	Current physical time
CNTP_CVAL_EL0	Target time
CNTP_TVAL_EL0	Relative interval
CNTP_CTL_EL0	Control register

2. Timer Compare Concept

The timer continuously performs:

Current Count

vs

Target Count

When:

Current Count >= Target Count

Timer expires.

Example

Parameter	Value
Current Count	1,000,000
Target Count	1,024,000

When:

Counter reaches 1,024,000

the timer expires.

3. Timer Expiry Flow

Step	Action
1	Counter advances
2	Compare logic checks target
3	Match occurs
4	Timer expires
5	Interrupt event generated

4. Physical Timer Registers

CNTP_CVAL_EL0

Stores:

Absolute target count

CNTP_TVAL_EL0

Stores:

Relative interval

CNTP_CTL_EL0

Contains:

Field	Purpose
ENABLE	Enable timer
IMASK	Mask interrupt
ISTATUS	Timer expired indication

5. Physical Timer Relationship

Component	Purpose
System Counter	Source of time
Physical Counter	Current time value
Physical Timer	Event generator
Interrupt	Notification mechanism

Key Takeaways

Concept	Description
Timer Source	Physical Counter
Compare Value	Target count
Expiry Condition	Current Count >= Target Count
Event	Interrupt generation

Knowledge Check

Q1

What does the Physical Timer use as its time source?

Q2

When does the timer expire?

Q3

Which register contains the absolute compare value?

Q4

Which register contains timer control bits?

Answer Key

Question	Answer
Q1	Physical Counter
Q2	Current Count >= Target Count
Q3	CNTP_CVAL_EL0
Q4	CNTP_CTL_EL0